A Multi-Hazard Assessment of 2,696 US Data Centers

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Background

- AI and cloud computing have concentrated US computing capacity into a few hundred sites.
- Those sites face earthquakes, wildfire and flooding, but no open dataset records where they physically stand.
- Existing sources are proprietary, aggregated above the building level, or unverified.
- We measured exposure: whether a facility sits in a hazard zone. We did not model damage, so nothing here is a loss estimate.

Bounds on these results

- Coverage is incomplete. Open sources list 2,696 US facilities where commercial directories list several thousand, and mapping density varies by state.
- Hail and wind reports rise with observer density (Spearman 0.39 and 0.34 within state), so we excluded both.
- Power capacity is unrecorded, so we count facilities, never megawatts.
- 48 facilities shown as facing no hazard sit outside FEMA's mapped extent, so 35% is a floor.
- Across 500 relocations, 1,681 sites never changed wildfire class.

Materials and Methods

We merged OpenStreetMap, PeeringDB and Wikidata into one row per site. Records within 60 m with matching fuzzy names are unioned, campus siblings are kept apart, and every merge is logged.

We then graded each coordinate against FEMA USA Structures building footprints and sampled hazards in an equal-area projection:

Earthquake	USGS ASCE 7-22, MCE_G PGA, site class BC
Wildfire	USFS Hazard Potential 270 m, max in 2.4 km
Flood	FEMA NFHL layer 28, SFHA flag
Water stress	WRI Aqueduct 4.0 (a resource limit, not a hazard)
Uncertainty	500 draws, sigma 10-500 m by coordinate tier

A facility counts as exposed if it lies within 2.4 km of land rated High or Very High for wildfire, inside a FEMA Special Flood Hazard Area, or at peak ground acceleration of 0.30 g or above (2% chance in 50 years). Where a hazard layer has no value we record it as unknown rather than counting the site as safe.

Results

65%	15	10	2,696
face no mapped hazard	states below 10% exposed	states above 60%	facilities mapped

Where a facility sits predicts its hazard exposure far better than anything about the facility itself.

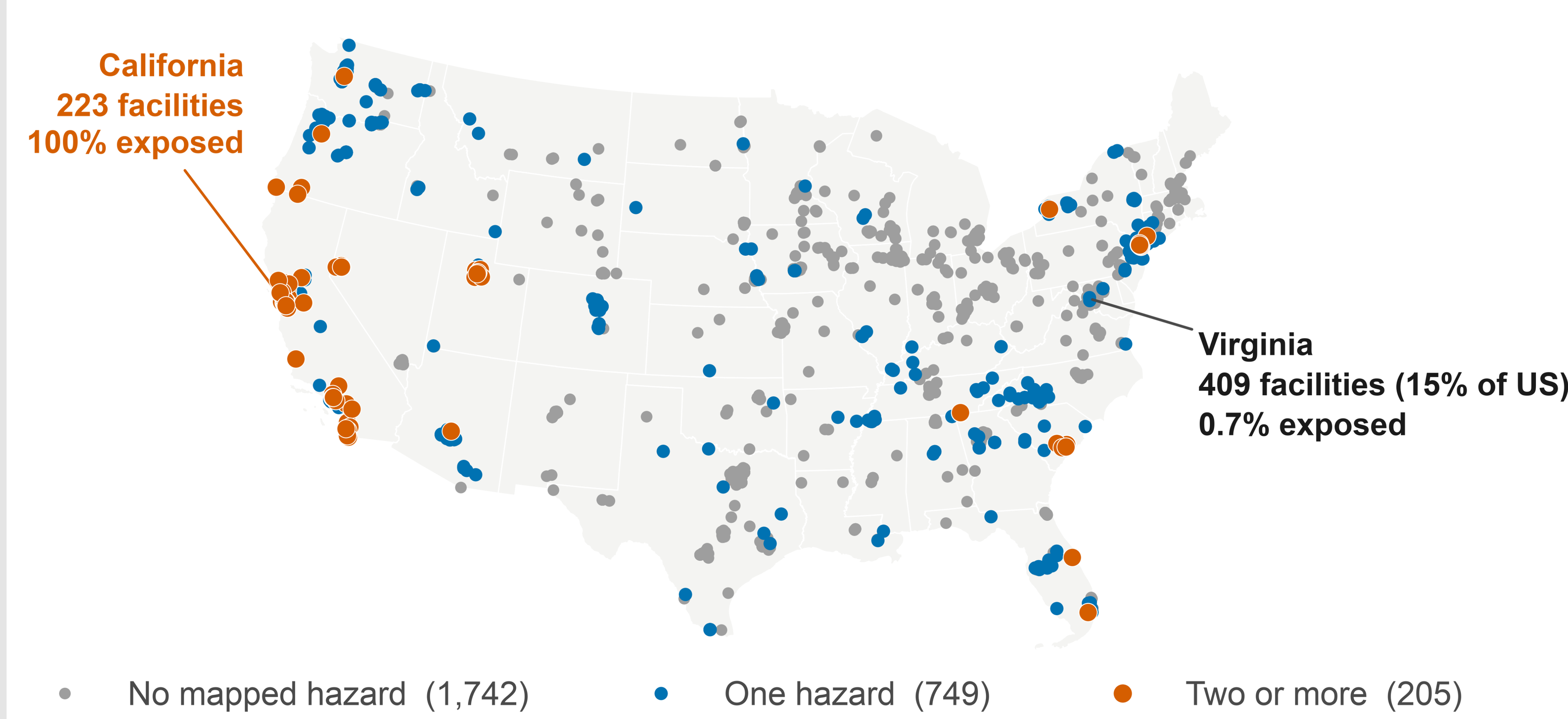

Fig 1. Two thirds of US data centers face no mapped hazard. The exposed ones cluster in the West and the Northeast.

Which hazard dominates changes by region. New Jersey reaches 100% entirely through wildfire and 0% through earthquake. California is 96% earthquake. Nationally wildfire drives 646 facilities, earthquake 448, flood only 71.

Water stress is the one hazard that reaches Virginia. 31% of its facilities sit in high-stress basins, against 0.7% for mapped hazards.

At a 5 km wildfire buffer the national count rises to 953, more than one third of the fleet.

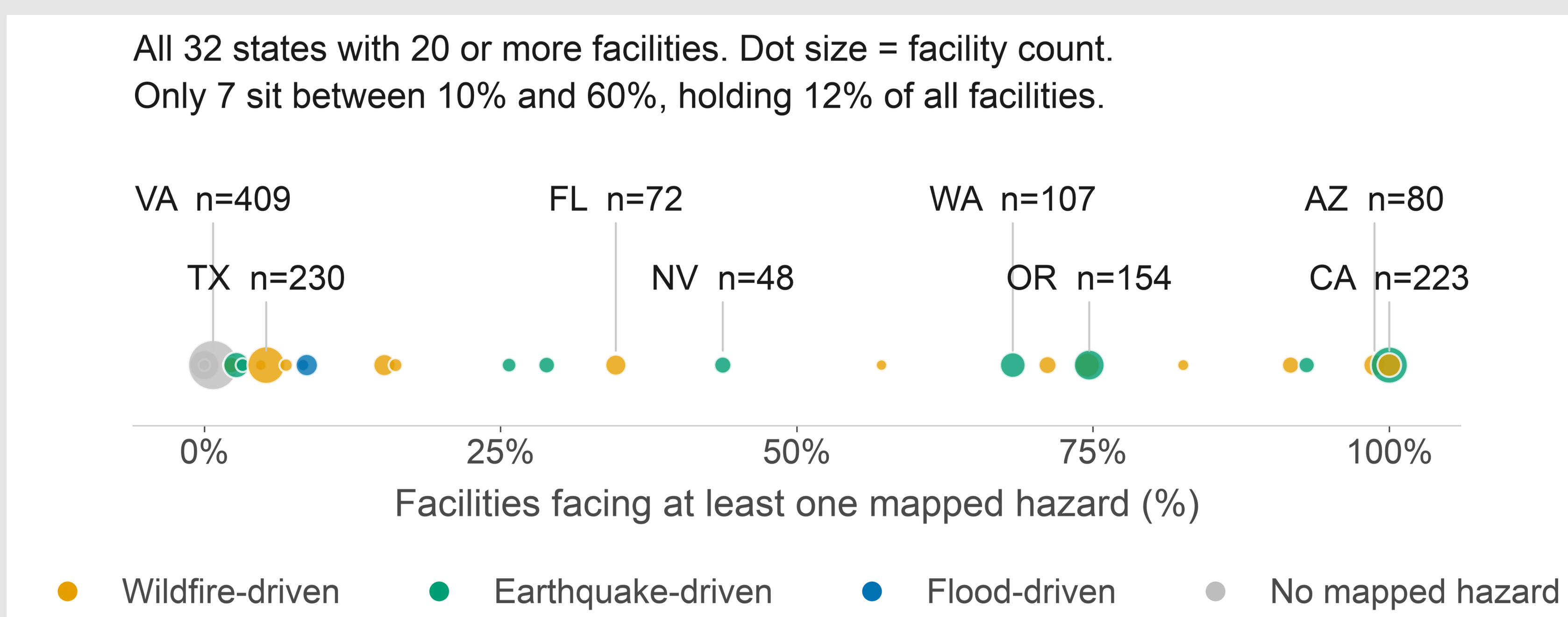

Fig 2. Every state with 20 or more facilities. 25 of 32 sit below 10% or above 60%, and colour shows which hazard drives each one.

Conclusions

- The national 35% figure describes no real state.
- Most sit in low-exposure regions, but we did not test why. Power price and fiber routes are likelier drivers than hazard avoidance.
- Two states can both read 100% and face different hazards, so a combined index must keep them separate.
- Next: fragility curves to turn exposure into estimated loss.

Two checks that came back negative

Measuring hazard across the whole footprint rather than at one point changed the answer for 29 of the 326 facilities where the two could be compared. Building height showed no association with exposure once state was held constant.

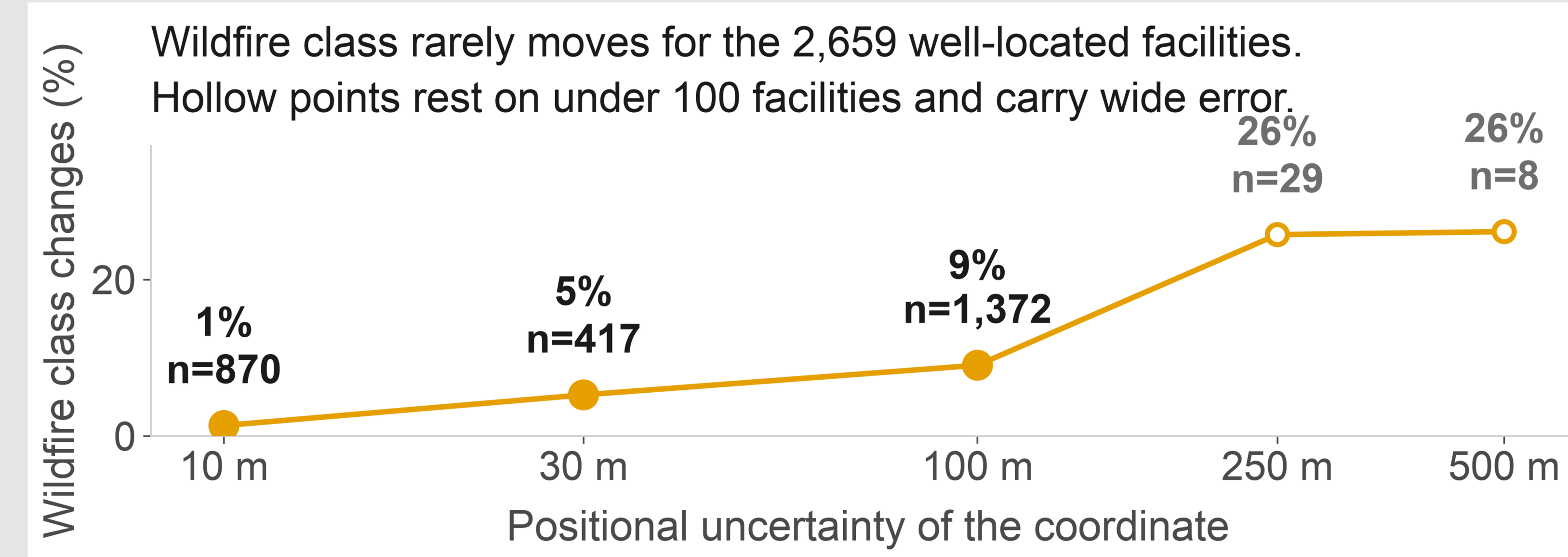

Fig 3. We re-ran the analysis with simulated location error.

Major Citations

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FEMA and ORNL (2024) USA Structures. gis.fema.gov

WRI (2023) Aqueduct 4.0 Water Risk Atlas.

Acknowledgements

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Data and code: github.com/aviangirekula/datacenter-dataset